MOTS-c

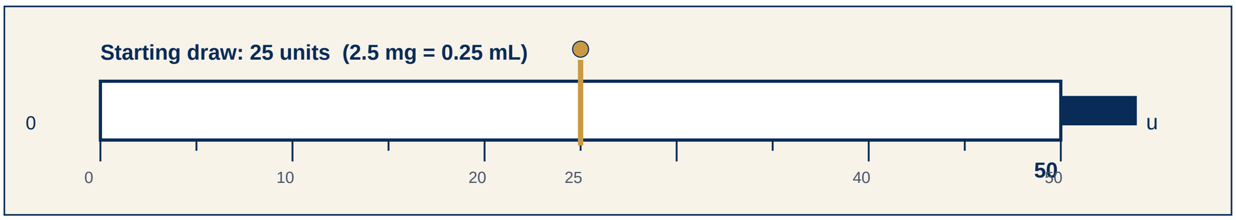
Information sheet written for the 20 mg vial TrueHold holds in stock. Not a generic internet protocol.

THIS VIAL — STARTING DOSE (read this first)

MOTS-c · 20 mg lyophilized · mix with 2 mL bacteriostatic water · 10 mg/mL · 1 unit = 100 mcg

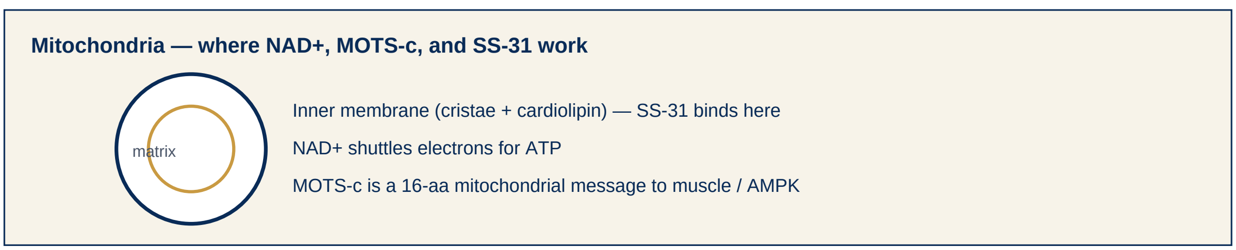
Start: draw **25 units** on a U-100 syringe (**0.25 mL** = **2.5 mg**). On a 0.5 mL (50-unit) U-100 syringe, draw to the 25 mark.

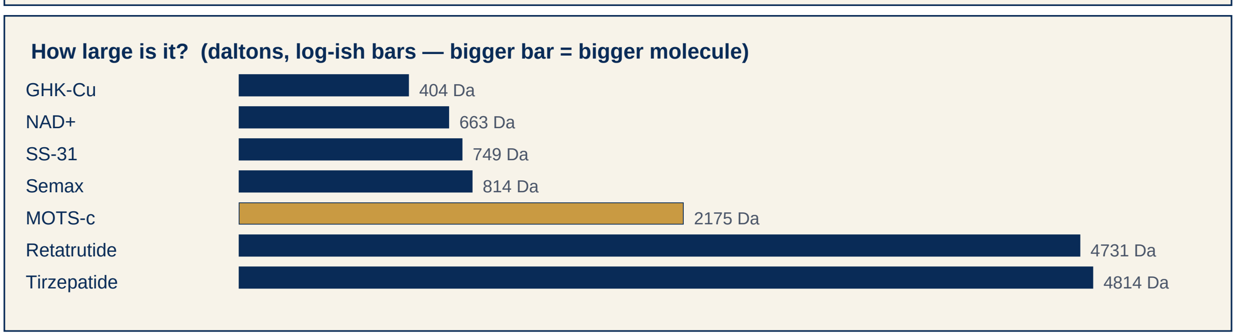
Route: Subcutaneous (belly or upper arm)
Rhythm: Three days per week (for example Mon/Wed/Fri) for 4–8 weeks, then a 2–4 week break.



1. What it is — the molecule, how large it is

MOTS-c (mitochondrial open reading frame of the 12S rRNA type-c) is a 16-amino-acid peptide encoded in mitochondrial DNA, not nuclear DNA. Sequence MRWQEMGYIFYPRKLR. Molecular weight about 2,175 daltons. TrueHold stocks a 20 mg lyophilized vial.





2. How it works in the body

Cells release MOTS-c as a mitochondrial signal. In muscle it is discussed as an AMPK-linked, exercise-mimetic message: more glucose uptake, more fatty-acid use, better insulin sensitivity in animal models. It is not a GLP-1 agonist and not NAD+.

3. Testing data — animal

Lee and colleagues showed MOTS-c prevented diet-induced obesity and insulin resistance in mice at roughly 0.5 mg/kg. Exercise raises endogenous MOTS-c in human muscle; that is observational biology, not a dosing licence.

4. Testing data — human, what is expected, hopes

No completed Phase 3. A Phase 2a subcutaneous study in prediabetes (NCT07505745) is registered. FDA compounding reviews have treated MOTS-c as poorly characterized for 503A bulk lists. Research-handling writeups use about 5–10 mg three times weekly; this 20 mg vial starts at 2.5 mg (25 units) then 5 mg (50 units, a full 0.5 mL syringe).

- Steadier energy and training output
- Insulin-sensitivity and body-composition research themes
- An exercise-linked mitochondrial signal without being a stimulant

5. Have people hurt themselves? Risks

There is no mature human safety database. Theoretical risks: low blood sugar if you already use insulin or intense fasting, unknown immunogenicity, unknown long-term organ effects. Injection-site irritation is the practical near-term issue. Do not treat mouse obesity papers as proof it is safe for you.

6. FDA — approved or not

NOT FDA-APPROVED for any indication. Not a legal compounded drug on the FDA bulk list as of the 2026 peptide reviews. TrueHold's 20 mg vial is research material.

7. Reconstitution for this exact vial

Check the label: it must say MOTS-c and 20 mg powder in a standard ~3 mL glass vial. You need bacteriostatic water (BAC), alcohol swabs, and U-100 insulin syringes (0.5 mL / 50-unit for every starting draw on this sheet; 1 mL / 100-unit if a later step exceeds 50 units).

- Wash hands. Wipe the rubber top with an alcohol swab.
- Draw **2 mL** of BAC into a mixing syringe.
- Insert through the stopper. Inject slowly down the inside glass wall — not straight onto the powder.
- Gently roll or swirl until clear. Do not shake.
- Wipe the top again. Write today's date on the vial. Refrigerate 36–46°F. Use within 28 days.
- Mix and store in this vial only. Do not transfer.

8. Loading the 0.5 mL syringe — schedule for this vial

U-100 math is the same on a 0.5 mL or 1 mL syringe: 100 units = 1.00 mL, so 50 units = 0.50 mL (a completely full 0.5 mL syringe). Starting draws on this sheet all fit 0.5 mL. The green row is the start.

When	Units (U-100)	mg	mL	0.5 mL syringe	How often
Weeks 1–2 (start here)	25	2.5	0.25	Draw to 25	Three days per week
Weeks 3–8 (if tolerated)	50	5	0.50	Draw to 50	Three days per week

- 20 mg in 2 mL → 10 mg/mL. 25 units = 2.5 mg; 50 units = 5.0 mg (a full 0.5 mL syringe).
- 5–10 mg three times weekly is the range scaled from published mouse work (Lee et al. ~0.5 mg/kg) and used in research-handling writeups. There is no FDA-approved human dose.
- At 25 units three times weekly one mixed vial lasts a little over 2 weeks.

Fulfillment

- Prep and local delivery: Las Vegas residents only.
- Shipping: dry (lyophilized) vials only — we do not ship mixed liquid.
- Interest orders are confirmed by phone before payment.

Working with the team

- Share a phone number so the team can call.

- Confirm Las Vegas residency for prep and local delivery.
- Complete required documentation with the team.
- If anything on this sheet disagrees with your vial label or COA, stop and call before mixing.

Sources

- Lee C. et al., mitochondrial MOTS-c metabolic papers; Front. Endocrinol. reviews (2023)
- ClinicalTrials.gov NCT07505745 (Phase 2a, subcutaneous)
- 20 mg vial / 2 mL BAC: 25 units = 2.5 mg, 50 units = 5 mg
- 20 mg TrueHold vial / 2 mL BAC. Starting 2.5 mg and step to 5 mg stay inside the commonly cited 5–10 mg research window without emptying the vial in two shots.

Protocol note

TrueHold writes these sheets for the milligram vials we hold. Mix volume and unit marks were double-checked so milligrams, milliliters, and U-100 units agree. This is educational research information, not a prescription and not medical advice. The Telegram bot does not dose in chat — read this PDF. For an individualized plan, use /schedule. Research use only.